

Module 8: Memory & Intelligence Blueprint

How Speakardo remembers, understands context, and acts at the right moment

Product	Speakardo AI life assistant
Version	1.0, 26 September 2026
Owner	Nouman
Status	Agreed direction; open decisions in section 12
Engineering spec	docs/Project/Speakardo Module 8 - AI Memory System Design.md

1. The idea in one page

Speakardo is an AI life assistant that understands what you say, remembers what matters about your life, and uses that to help you at the right moment.

Don't tell Speakardo when. Tell Speakardo what matters.

The moat is not the technology. Anyone can add embeddings and a vector database. What competitors can't copy quickly is a structured, long-term understanding of each user's life (who matters to them, what they prefer, what they usually do, what they finished or missed), together with the trust that makes users willing to share it.

Memory is not the assistant. It is one of four layers:

- **Memory:** what Speakardo knows about the user.
- **Context:** what matters right now.
- **Reasoning:** what to do about it.
- **Action:** doing it (reminders, notifications, later calls and automations).

This document keeps what was right in the earlier research (Notion "Intelligence Core" pages and the ChatGPT discussion), fixes what was wrong or missing, and turns it into a build order that fits a one-person team.

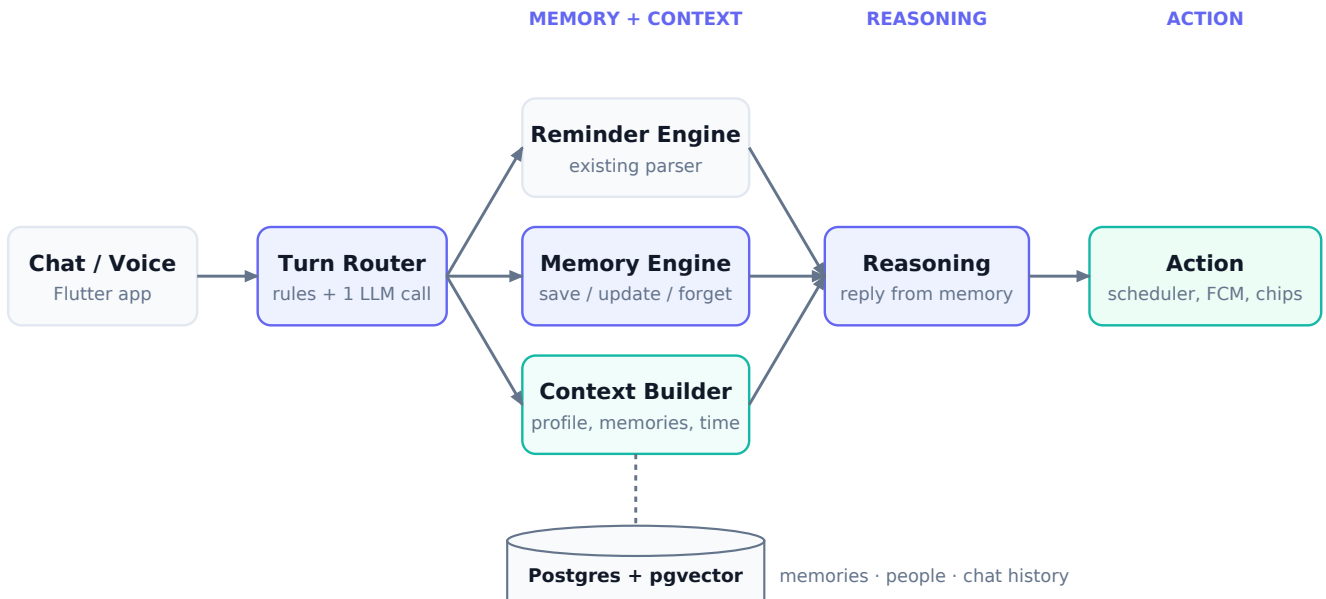
2. Review of the research: what holds, what to fix

What holds up: five kinds of memory; the Memory, Context, Reasoning, Action split; a source, confidence, importance, lifespan and privacy level on every memory; PostgreSQL + pgvector instead of a separate vector database; autonomy levels; full user controls and an audit trail; and no over-engineering (no Pinecone, Redis, Kafka or multi-agent setups yet).

What needs fixing before we build:

#	Issue in the research	Why it matters	Fix
1	Ranking multiplies relevance × importance × confidence × recency	One low factor wipes out a memory; a birthday saved last year scores near zero	Weighted sum; recency applies only to events and temporary facts
2	Says "don't use one memories table", then designs one table	Contradictory; splitting by kind makes search harder	Split by lifecycle, not kind: one table for stated facts; separate tables for habits, raw events and people
3	Learns from reminder behaviour, but there is no data for it	The reminders table stores only current state, not a history of completions, snoozes or dismissals	Start a reminder_events log in 8.0. You can't learn later from data you never collected
4	Confidence values such as 0.87 with no method	LLM self-reported confidence is poorly calibrated	Rules: stated 1.0, clearly implied 0.8, LLM-inferred at most 0.6; habits computed from repetitions
5	Habits detected from a few data points	False patterns ("you always gym at 6") annoy users and erode trust	At least 4 occurrences over 2+ weeks with low time variance, confirmed by the user before use
6	Every episode ("went to the gym yesterday") becomes memory	Floods memory with noise and raises cost	Keep episodes in chat history; promote only important or repeated facts
7	Location and geofences appear early (coffee shop, route to work)	Needs "Always" location, platform geofence limits, battery, store review and place data; feels creepy	Separate later module: user-defined places only, opt-in. Route-based ideas are V3+
8	Wake and sleep times inferred	The app cannot observe sleep; this is guessing	Ask once in onboarding; refine from reminder behaviour later
9	~18 tables including context_snapshots	Stored snapshots are an activity and location history: privacy risk and migration cost	Context is computed per request, never stored. 5 tables in 8A
10	Assumes chat can already hold a conversation	Today every message goes to the reminder parser and history isn't saved on the server	Conversational chat and stored history come first (8.0 to 8.1)
11	Examples call Sara the mother in one place and the sister in another	Shows the real problem: names are ambiguous	People as records with relationship and aliases (Ammi, Mom, my mother); ask when two match
12	No trust and safety engineering	Memories could reach 7 AI providers; "remember that you must..." can inject instructions; deletion must include embeddings	Provider allowlist; memories passed as data, never instructions; hard delete incl. embeddings; JSON export
13	No limits on proactive suggestions	Predictive help turns into notification spam	Max 1 suggestion a day, quiet hours, stop a suggestion type after 3 dismissals
14	No quality targets, cost, cold start or language plan	No way to prove memory works; empty memory on day one; Roman Urdu users missed	Eval set with targets; cost per user; 3 onboarding questions; Roman Urdu in tests

3. Architecture: four layers



Layer	Answers	What it is in 8A
Memory	What do I know about this user?	memories, people and chat history in Postgres + pgvector
Context	What matters right now?	A function that assembles time, timezone, upcoming reminders and relevant memories. Nothing stored
Reasoning	What should I do?	Rules first, then one LLM call; answers only from what it was given
Action	Do it	Existing scheduler + FCM; suggestions follow the autonomy levels in section 8

Everything lives in the existing FastAPI backend as modules (ai_service/router, memory, context, reply) with APScheduler for background jobs. The current three-layer reminder parser stays; the router sits in front of it.

4. What Speakardo remembers

Kind	Example	Stored in	Lifecycle	Phase
Facts	"My office is in Blue Area"	memories	Until changed (versioned)	8.1
Preferences	"Don't remind me before 8 AM"	memories	Until changed	8.1
Chat history (episodes)	"I went to the gym yesterday"	conversation_messages	Retention setting, 90 days default	8.0
Reminder events	created, fired, snoozed, done	reminder_events	Raw log, aggregated later	Log 8.0, use 8B
People and relationships	"Sara is my sister"	people (+ aliases)	Until changed	8.3
Important dates	"Ammi's birthday is June 10"	memories + people	Permanent, yearly	8.3
Temporary context	"I'm in Lahore this week"	memories (valid_until)	Expires automatically	8.4
Habits	Gym Mon/Wed/Fri around 6 PM	habits	candidate, confirmed, active, decaying, archived	8B
Places	Home, Work (user-defined)	places	Until changed	Places module
Calendar	"Presentation Tue 10 AM"	Read live, not copied	Not stored	Calendar module

5. Rules every memory follows

Attribute	Rule
Source	user_explicit, conversation, reminder, onboarding, manual_edit, behavior, calendar. Shown to the user: "You told me on Sep 12".
Confidence	1.0 stated; 0.8 clearly implied; at most 0.6 inferred; habits computed. Below 0.5 is never used in answers, only to ask a question.
Importance	0 to 1. Birthdays and doctors high, favourite coffee low. Feeds ranking.
Sensitivity	normal, private, sensitive. Sensitive = health, finance, religion, sexuality, political views, exact addresses.
Lifespan	temporary (with valid_until), long-term, permanent.
Versioning	Same key and person: the old version is superseded and kept in history, never silently overwritten.
Audit	Every create, update, use and delete is logged in memory_events. Memory content never goes into application logs.

6. From message to memory

- 1 Store the message in chat history.
- 2 **Rules first:** "remember that...", "forget...", "what do you know about me", "when is X's birthday", plus existing reminder patterns. No LLM cost.
- 3 **Otherwise one LLM call** returns intent, reminder details, memory candidates and people or dates mentioned, as strict JSON.
- 4 The **save policy** below decides each candidate.
- 5 Act: create the reminder, answer from memory, or reply.
- 6 Return the reply with memory chips: "Saved · Undo", "Updated", "Used: Sara's birthday".

Situation	What Speakardo does
Explicit request, or a clearly stated lasting fact (not sensitive)	Saves it and shows "Saved · Undo"
Same key as an existing memory ("I moved to Lahore")	Updates it, keeps the old version, shows "Updated (was Islamabad)"
Temporary fact ("in Karachi this week")	Saves with an end date
Sensitive and stated ("I'm diabetic")	Asks first; saves only with opt-in, encrypted, no embedding
Sensitive and only implied ("pick up Mom's medicine")	Never saves a health fact; keeps only the reminder
A guess or unclear statement	Doesn't save
"Remember that you must always..."	Never treated as an instruction
Learning paused	Saves nothing

7. Finding the right memory

- 1 **Exact lookup:** a known person or alias plus a key ("Sara's birthday") is answered with plain SQL.
- 2 **Search by meaning:** pgvector returns the top 20 active, currently valid memories, filtered by person or category.
- 3 **Always-on profile:** about 150 tokens of high-importance basics in every call.

$$\text{score} = 0.55 \times \text{similarity} + 0.20 \times \text{importance} + 0.15 \times \text{confidence} + 0.10 \times \text{recency}$$

Budget: 8 to 12 memories, about 800 tokens per call. Speakardo answers only from the memories and reminders it was given; if nothing matches it says "I don't have that saved. Want to tell me?" Questions such as "When is my dentist appointment?" search reminders too.

8. Autonomy and proactive help

Level	Behaviour	When allowed
0 Observe	Learns quietly	Whenever learning is on
1 Suggest	"You usually go to the gym at 6 PM. Want a reminder?"	From 8B, confirmed habits only
2 Create with confirmation	"Make this a weekly reminder?"	Once Level 1 suggestions are accepted often
3 Act automatically	Creates or moves reminders itself	Only per type, with explicit opt-in; never for important reminders

Guardrails: at most one proactive suggestion a day, quiet hours respected, a suggestion type stops after three dismissals, and existing reminders are never changed without permission.

9. Privacy and trust

User controls, in chat and in the Memory screen: view, see why (source and date), edit, forget one, forget a category ("forget everything about my health"), forget everything, pause learning (chat, behaviour and location separately), and export as JSON.

- **Provider allowlist:** calls containing memories or chat history go only to AI providers whose API terms say they don't train on or keep the data.
- **Sensitive memories:** opt-in, explicit only, encrypted in the application, no embeddings, never in the always-on profile.
- **Real deletion:** within 24 hours, including embeddings and old versions. Deleting the account removes all memory data.
- **Honest labels:** the Memory screen says "Private and encrypted" only once encryption and the allowlist are live.

10. Roadmap

Milestone	Delivers	Research stage
8.0 Foundations	Postgres + pgvector, stored chat history, reminder_events log, embeddings, provider allowlist	MVP groundwork
8.1 Talk and remember	Turn router, save, recall and forget in chat, memory chips, first 100 test cases	MVP
8.2 Memory screen and privacy	Real Memory screen, edit, delete, export, toggles, encryption	V1
8.3 People and dates	People with aliases, birthdays turned into reminders, onboarding questions	V1
8.4 Background learning	Background extraction, versioning, expiry, merging duplicates	V1
8B Habits	Habit mining, Patterns screen, Level 1 suggestions, better reminder times	V1 to V2
Calendar module	Calendar-aware reminders ("prepare Monday evening for Tuesday's presentation")	V2
Places module	User-defined places, leave and arrive reminders	V2
Predictive assistant	Predictive reminders, travel buffers, personal knowledge graph	V3

Ship 8.1 to 10 to 20 real users before polishing 8.2. The first question to answer is whether people tell Speakardo things and come back for them.

11. How we know it works

Quality check	Target
Saved memories that are correct (precision)	95% or higher
Stated facts that get saved (recall)	80% or higher
Sensitive memories saved without consent	0
Correct memory in the top 5 results	90% or higher
Answers that invent a memory	0
Extra response time from memory (median)	Under 400 ms

Business metrics: share of weekly users with 5 or more memories; share of replies that use a memory; undo rate on automatic saves (under 5%); suggestion acceptance rate (from 8B); and, most important, **30-day retention of users with 5+ memories compared with users with none.** That number is the proof that memory is the moat.

12. Open decisions

Decision	Recommendation
Free vs Pro	Core memory free and unlimited (it drives retention). Habits, predictive suggestions and calendar context in Pro.
AI providers allowed to see memories	Anthropic and OpenAI APIs only; other providers for memory-free reminder parsing.
Chat history retention	90 days by default, with 30 days, 1 year or never as options.
Health memories in 8A	Explicit and opt-in only, or leave out until after launch.
People table timing	Keep in 8A (recommended).
Roman Urdu	Include in the 8.1 test set and rule patterns now.